

① Job Sequencing with Deadlines

It is a scheduling problem where we aim to maximize profit by scheduling jobs within their deadlines.

- There are n jobs, each having a deadline $d_i \geq 0$ & Profit $p_i \geq 0$. We need one unit of time to process each job we can do at most one job each time. The p_i is earned if job i is completed by its deadline.

feasible sol: subset of jobs j such that each job is completed by its deadline.

Optimal sol: feasible solⁿ with maximum profit value.

$$n=5; (p_1, p_2, p_3, p_4, p_5) = (50, 35, 20, 10, 1)$$

$$(d_1, d_2, d_3, d_4, d_5) = (4, 1, 2, 2, 3)$$

Step 1:

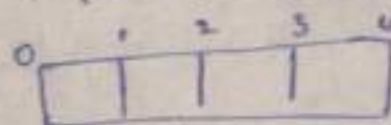
Sort all jobs in decreasing order of Profit.

$$(p_1, p_2, p_3, p_4, p_5) = (50, 35, 20, 10, 1)$$

$$(d_1, d_2, d_3, d_4, d_5) = (4, 1, 2, 2, 3)$$

Step 2:

Draw Gantt chart with max deadline = 4

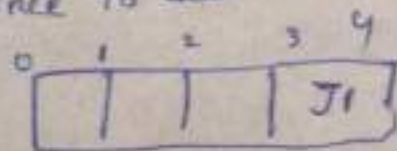


• we take each job one by one in the order they appear in step 1.

Step 3:

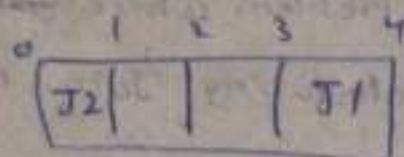
we take Job J_1

since its deadline is 4. we place it just before 4.



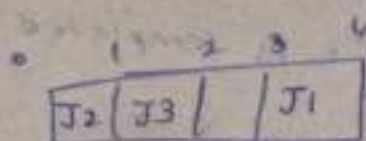
step 4:

- we take Job J₂
- since its deadline is 1 we place it just before 1.



step 5:

- we take Job J₃
- Its deadline is 2. we place it just before 2.



step 6:

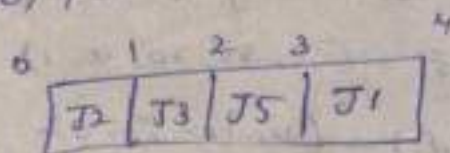
we take Job J₄.

- Its deadline is 2. there are no empty slots before 2 so we cannot place it.

step 7:

Next Job is J₅

- Its deadline is 3. there is an empty slot just before 3. So, place it there.



$$\text{Profit} = \{J_2, J_3, J_5, J_1\}$$

$$= \{P_2, P_3, P_5, P_1\}$$

$$= 35 + 20 + 1 + 50 = 106$$

③ Activity Selection Problem

There are n activities with start & finish time.
select maxm no. of activities that can be performed
by a single person, assuming that a person can only
work on a single activity at a time.

$$S = (A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9, A_{10})$$

$$S_i = (1, 2, 3, 4, 7, 8, 9, 9, 11, 12)$$

$$F_i = (3, 5, 4, 7, 10, 9, 11, 13, 12, 14)$$

Step 1: sort activities in ascending order of their finishes

activity	A ₁	A ₃	A ₂	A ₄	A ₆	A ₅	A ₇	A ₉	A ₈	A ₁₀
start	1	3	2	4	8	7	9	11	9	12
Finish	3	4	5	7	9	10	11	12	13	14

Step 2: A₁ is selected ✓ (move start) (don't change finish)

Step 3: start[A₃] > end[A₁]

A₃ accepted ✓ [move start & finish by 1]

Step 4: start[A₉] < end[A₃]

rejected [move start]

Step 5: start[A₄] > end[A₃]

accepted A₄ ✓ [move start]

Step 6: start[A₆] > end[A₄]

A₆ accepted ✓

[set finish before the slot of finish]

[0 4]

[0 4]
[0 4]

step 7: start $[A_5] < \text{end } [A_5]$
rejected $\times$

step 8: start $[A_7] < \text{end } [A_5]$
rejected $\times$

step 9: start $[A_9] > \text{end } [A_5]$
 A_9 accepted $\checkmark$

step 10: start $[A_3] < \text{end } [A_9]$
rejected $\times$

step 11: start $[A_{10}] > \text{end } [A_9]$
 A_{10} accepted $\checkmark$

Selected activities: $A_1, A_3, A_6, A_8, A_9, A_{10}, A_{11}$

(c) $(a_1, a_2, a_3, a_4): (\text{do, if int, while})$

$$p(1:4) = (3, 3, 1, 1)$$

$$q(0:4) = (2, 3, 1, 1, 1)$$

$$A) w(i, j) = w(i, j-1) + p[j] + w[i]$$

$$c(i, j) = \min_{1 \leq k \leq j} \{ c(i, k-1) + c(k, j) \} + w(i, j)$$

$$r(i, j) = k \quad (\text{for which 'k' value min value is produced})$$

Initial conditions

$$w(1, i) = q_1(i)$$

$$c(1, i) = 0$$

$$r(i, i) = 0$$

$i=0$:

$$\begin{array}{l|l|l} w(0,0) = q_0 = 2 & w(1,1) = q_1 = 3 & w(2,2) = q_2 = 1 \\ c(0,0) = 0 & c(1,1) = 0 & c(2,2) = 0 \\ r(0,0) = 0 & r(1,1) = 0 & r(2,2) = 0 \end{array}$$

$$\begin{array}{l|l} w(3,3) = q_3 = 1 & w(4,4) = q_4 = 1 \\ r(3,3) = 0 & r(4,4) = 0 \\ r(4,4) = 0 & c(4,4) = 0 \end{array}$$

	0	1	2	3	4
$j-i=0$	$w_{00} = 2$ $c_{00} = 0$ $r_{00} = 0$	$w_{01} = 3$ $c_{01} = 0$ $r_{01} = 0$	$w_{02} = 7$ $c_{02} = 0$ $r_{02} = 0$	$w_{03} = 1$ $c_{03} = 0$ $r_{03} = 0$	$w_{04} = 1$ $c_{04} = 0$ $r_{04} = 0$
$j-i=1$	$w_{10} = 3$ $c_{10} = 0$ $r_{10} = 1$	$w_{11} = 7$ $c_{11} = 7$ $r_{11} = 2$	$w_{12} = 3$ $c_{12} = 3$ $r_{12} = 3$	$w_{13} = 3$ $c_{13} = 3$ $r_{13} = 4$	
$j-i=2$	$w_{20} = 12$ $c_{20} = 19$ $r_{20} = 1$	$w_{21} = 9$ $c_{21} = 12$ $r_{21} = -2$	$w_{22} = 5$ $c_{22} = 8$ $r_{22} = 3$		
$j-i=3$	$w_{30} = 14$ $c_{30} = 25$ $r_{30} = 2$	$w_{31} = 11$ $c_{31} = 19$ $r_{31} = 2$			
$j-i=4$	$w_{40} = 16$ $c_{40} = 3$ $r_{40} = 2$				

$i=0$
 $w(0,0) = q_0 = 2$
 $c(0,0) = 0$
 $r(0,0) = 0$

$i=1$
 $w(1,1) = q_1 = 3$
 $i=2$
 $w(2,2) = q_2 = 1$

$i=3$
 $w(3,3) = q_3 = 1$
 $i=4$
 $w(4,4) = q_4 = 1$

$$\begin{aligned}
 w(0,1) &= w(0,0-1) + P(1)q(1) \\
 &= w(0,0) + P(1) + q(1) \\
 &= 2 + 3 + 3 = 8
 \end{aligned}$$

$$w(1,3) = w(1,2+1) + P(3)q(3)$$

$$\begin{aligned}
 c(0,1) &= \min_{0 \leq k \leq 1} \{ c(0,1-k) + c(1,k) \} + w(0,1) \quad \leftarrow i=0, j=1 \\
 &= \min \{ c(0,0) + c(1,1) \} + w(0,1) \\
 &= \min \{ 0 + 0 \} + 8 = 8
 \end{aligned}$$

$$r(0,1) = K = 1$$

$$\begin{aligned}
 w(1,2) &= w(1,1) + P(2)q(2) \\
 &= 3 + 3 + 1 = 7
 \end{aligned}$$

$$r(1,2) = K = 2$$

$$\begin{aligned}
 c(1,2) &= \min_{1 \leq k \leq 2} \{ c(1,2-k) + c(2,k) \} + w(1,2) \\
 &= \min \{ 0 + 0 \} + 7 = 7
 \end{aligned}$$

$$w(2,3) = w(2,2) + p[3] + q[3]$$

$$= 1 + 1 + 1 = 3$$

$$c(2,3) = \min_{2 \leq k \leq 3} \{ c(2,2) + c(k,3) \} + w(2,3)$$

$$= \min \{ 0 + 0 \} + 3 = 3$$

$$r(2,3) = k$$

$$= 3$$

$$w(3,4) = w(3,3) + p[4] + q[4]$$

$$= 1 + 1 + 1 = 3$$

$$c(3,4) = \min_{3 \leq k \leq 4} \{ c(3,3) + c(k,4) \} + w(3,4)$$

$$= \min \{ 0 + 0 \} + 3 = 3$$

$$r(3,4) = k$$

$$= 4$$

$$w(0,2) = w(0,1) + p[2] + q[2]$$

$$= 8 + 3 + 1 = 12$$

$$c(0,2) = \min_{0 \leq k \leq 2} \left\{ \begin{array}{l} c(0,0) + c(1,2) \\ c(0,1) + c(2,2) \end{array} \right\} + w(0,2)$$

$$= \min \left\{ \begin{array}{l} 0 + 7 \\ 8 + 0 \end{array} \right\} + 12$$

$$= 7 + 12 = 19$$

$$r(0,2) = k$$

$$= 1$$

$\left[\begin{array}{l} \text{when } k=1, \text{ it is producing } 7 \\ \text{when } k=2, \text{ " " " " } 8 \\ \therefore \text{ when } k=1, \text{ it is less} \end{array} \right]$

$$w(1,3) = w(1,2) + p[3] + q[3]$$

$$= 7 + 1 + 1 = 9$$

$$c(1,3) = \min_{1 \leq k \leq 3} \left\{ \begin{array}{l} c(1,2) + c(3,3) \\ c(1,1) + c(2,3) \end{array} \right\} + w(1,3)$$

$$= \min \left\{ \begin{array}{l} 0 + 3 \\ 7 + 0 \end{array} \right\} + 9$$

$$= 3 + 9 = 12$$

$$r(1,3) = k$$

$$= 2$$

$(\because \text{for } k=2, \text{ it is producing less i.e. } 3)$

$$w(2,4) = w(2,3) + p[4] + q[4]$$

$$= 3 + 1 + 1 = 5$$

$$c(2,4) = \min_{2 \leq k \leq 4} \left\{ \begin{array}{l} c(2,2) + c(4,4) \\ c(2,3) + c(4,4) \end{array} \right\} + w(2,4)$$

$$= \min \left\{ \begin{array}{l} 0 + 3 \\ 3 + 0 \end{array} \right\} + 5$$

$$= 3 + 5 = 8$$

$$r(2,4) = k$$

$$= 3$$

$(\because \text{for } k=3, \text{ it is producing less i.e. } 3)$
 $\text{in } k=3, (k=4)$

$$w(0,3) = w(0,2) + p[3] + q[3]$$

$$= 12 + 1 + 1 = 14$$

$$c(0,3) = \min_{0 \leq k \leq 3} \left\{ \begin{array}{l} c(0,0) + c(1,3) \\ c(0,1) + c(2,3) \\ c(0,2) + c(3,3) \end{array} \right\} + w(0,3)$$

$$= \min \left\{ \begin{array}{l} 0 + 12 \\ 8 + 3 \\ 19 + 0 \end{array} \right\} + 14$$

$$= 12 + 14 = 26$$

$$r(0,3) = k$$

$$= 2$$

$$w(1,4) = w(1,3) + p[4] + q[4]$$

$$= 9 + 1 + 1 = 11$$

$$c(1,4) = \min_{1 \leq k \leq 4} \left\{ \begin{array}{l} c(1,1) + c(2,4) \\ c(1,2) + c(3,4) \\ c(1,3) + c(4,4) \end{array} \right\} + w(1,4)$$

$$= \min \left\{ \begin{array}{l} 0 + 8 \\ 7 + 3 \\ 12 + 0 \end{array} \right\} + 11$$

$$= 7 + 11 = 18$$

$$r(1,4) = k$$

$$= 2$$

$$w(0,4) = w(0,3) + p[4] + r[4]$$

$$= 14 + 1 + 1 = 16$$

$$c(0,4) = \min_{\substack{0 \leq k < 4 \\ k=1 \\ k=2 \\ k=3 \\ k=4}} \left\{ \begin{array}{l} c(0,0) + c(1,4), \\ c(0,1) + c(2,4), \\ c(0,2) + c(3,4), \\ c(0,3) + c(4,4) \end{array} \right\} + w(0,4)$$

$$= \min \left\{ \begin{array}{l} 0+19, \\ 8+8, \\ 19+3, \\ 25+0 \end{array} \right\} + 16$$

$$= 16 + 8 = 24$$

$$r(0,4) = k$$

$$= 2$$

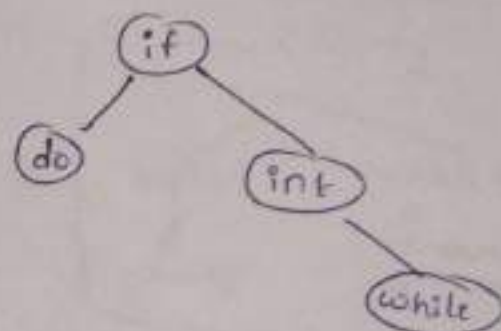
$$\text{let } r(i,j) = k$$

$$\begin{array}{cc} & \swarrow \quad \searrow \\ r(i, k-1) & \quad r(k, j) \end{array}$$

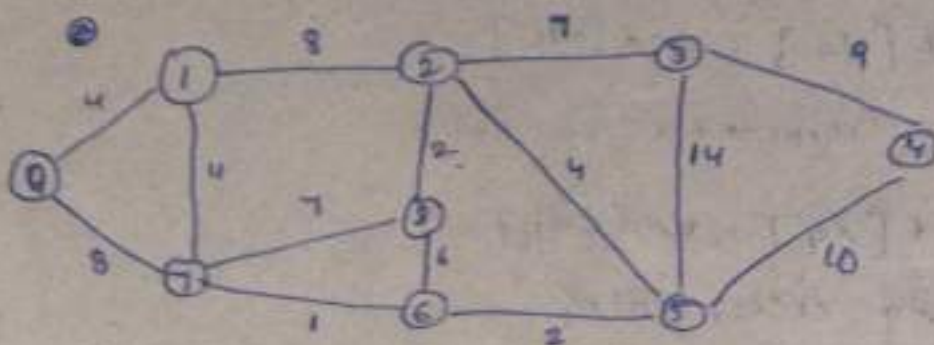
$\equiv$

$$\begin{array}{cc} & r(0,4)=2 \\ & \swarrow \quad \searrow \\ r(0,1)=1 & \quad r(2,4)=3 \\ \swarrow \quad \searrow & \quad \swarrow \quad \searrow \\ r(0,0)=0 \quad r(1,1)=0 & \quad r(2,2)=0 \quad r(3,4)=4 \\ & \quad \swarrow \quad \searrow \\ & \quad r(3,3)=0 \quad r(4,4)=0 \end{array}$$

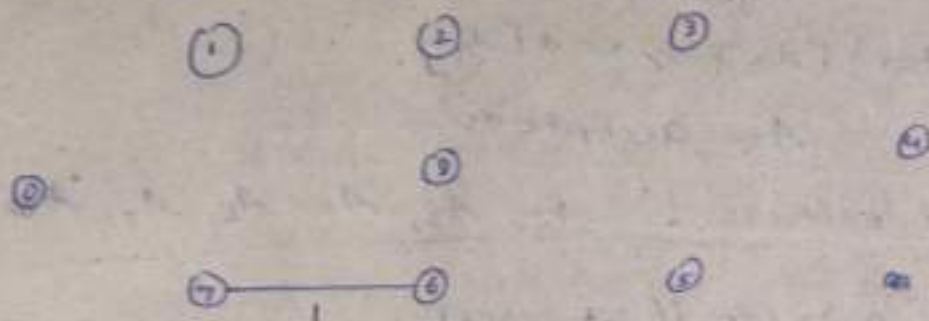
$\Rightarrow$



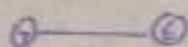
⑥ Kruskal's



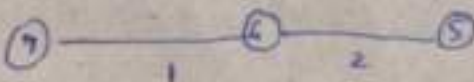
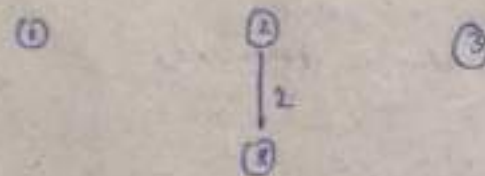
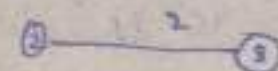
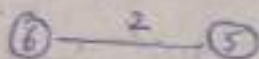
Step 1: Consider min cost edge = 1



min cost = 1

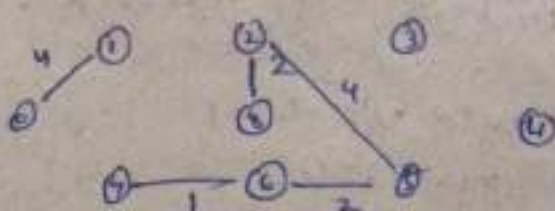
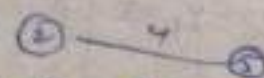


Step 2: Next min cost edge i.e



min cost = 1 + 2 + 2 + 4 + 4 = 13

Step 3: Next min cost edge i.e



min cost = 5 + 4 + 4 = 13

Step 4: Next min. cost edge is $(2) \xrightarrow{6} (6)$

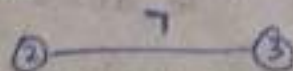
But it forms cycle.

So avoid this edge.

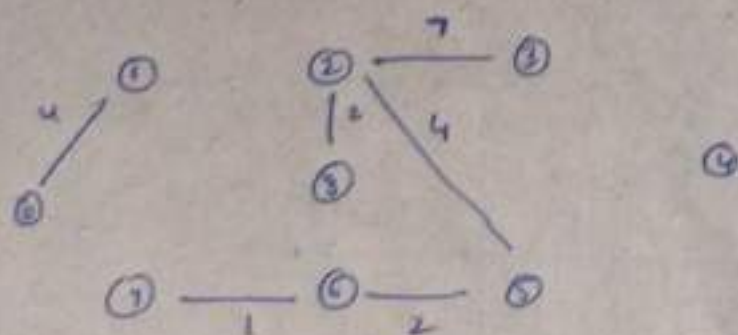
Next minimum edge $(7) \xrightarrow{7} (8)$

It also form cycle

So avoid this edge.



Cannot form cycle.

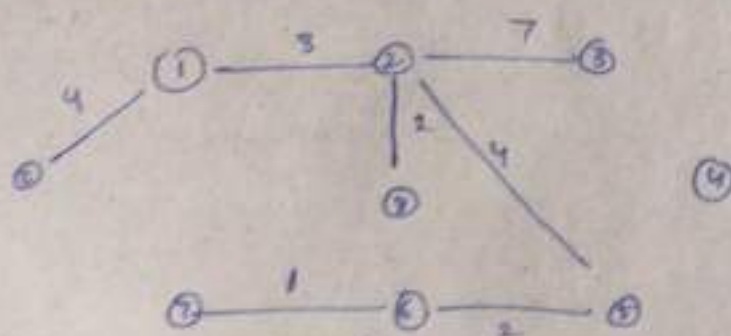


$$\text{min cost} = 14 + 7 = 21$$

Step 5: Next min edge $(1) \xrightarrow{8} (2)$

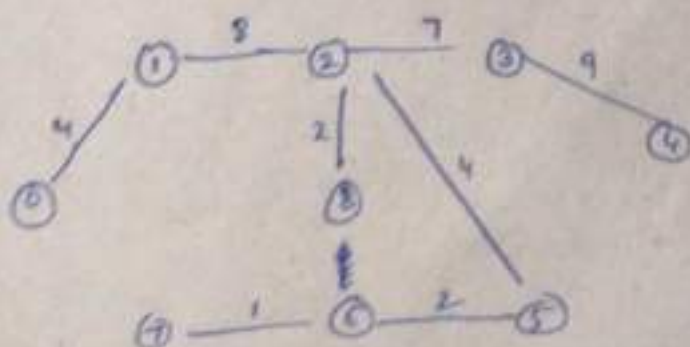
$(6) \xrightarrow{3} (7)$ will form cycle $(1) \xrightarrow{8} (2)$ is drawn.

So select any one.



$$\text{min cost} = 21 + 8 = 29$$

Step 6: Next min-edge = $(3) \xrightarrow{9} (4)$



$$\text{min cost} = 29 + 9 = 37$$

Step 7: Next min $(6) \xrightarrow{10} (4)$

It will form cycle
so avoid

Step 8: next min edge ① —⁴— ⑦

It forms cycle

So, avoid

Step 9: Next min edge ③ —¹⁴— ⑤

It forms cycle

So avoid

∴ cost of minimum Spanning Tree = 37